

4	(a)(i)	For same power transmission, high voltage results in low current With low current, less energy/power losses in transmission cables	B1 B1
	(a)(ii)	AC voltage is easily/efficiently change/step-up or step down using transformers	B1
	(b)(i)	To reduce power loss in the core due to eddy currents/induced current	M1
	(b)(ii)	Magnetic flux is phase with voltage/current in primary coil (by Faraday's law) Induced emf in secondary coil is proportional to rate of change of flux so voltage in primary coil not in phase with induced emf in secondary coil	M1 M1
	(c)	Rms voltage across load = $9.0 \times 8100 / 300 = 243 \text{ V}$ Peak voltage = $243 \times \sqrt{2} = 340 \text{ V}$	C1 A1